

SUBAR MAHDI – CV

PROFILE

Mechanical with Electrical Engineering student focused on developing real engineering capability through hands-on projects and team-based systems work. Currently building experience across Formula Student vehicle development, rocketry feed systems, applied research, and independent electromechanical projects. Former programmatic advertising lead with experience operating and optimising complex technical systems under pressure, now transitioning that systems thinking into engineering.

ENGINEERING EXPERIENCE

Mechanical and Electrical Team Member — Bristol Racing (Formula Student) Sep 2025 – Present

- › Working within the mechanical and electrical team on a scaled Formula Student RC vehicle
- › Supporting drivetrain and steering concepts, CAD assemblies, packaging, and wiring layout planning
- › Assisting with manufacturable design using laser-cut and 3D-printed components
- › Developing understanding of performance trade-offs, tolerances, reliability, and competition constraints

Systems Engineer (Feed Systems) — HyPower Rocket Society Sep 2025 – Present

- › Supporting design of pressurisation and propellant feed systems for competitive rocketry
- › Producing and interpreting P&IDs and CAD models for tanks, valves, and system routing
- › Learning subsystem integration, safety-driven design, and formal review processes
- › Building intuition around how design decisions affect testing, operations, and reliability

Junior Researcher — Academic Research Society (Engineering & Technology) Sep 2025 – Present

- › Conducting applied research into engineering and technology topics
- › Reviewing technical literature and producing concise technical summaries
- › Developing structured technical writing and analytical reasoning skills

PROJECTS

Smart Lamp — Personal Project Sep 2025 – Present

Ongoing electromechanical project focused on enclosure CAD, electronics integration, and iterative prototyping. Designing a custom housing, integrating Arduino-controlled lighting, refining wiring layouts, and learning practical packaging, robustness, and manufacturability through repeated build-test cycles.

Load-Bearing Single-Axis Linear Actuator — Personal Project Sep 2025 – Present

Ongoing mechanical project focused on designing a compact actuator capable of handling meaningful loads. Modelling load cases, producing CAD assemblies, and learning how tolerances, alignment, friction, and interfaces influence real-world performance during prototyping.

PROFESSIONAL EXPERIENCE

Programmatic Advertising Lead — Global Media Organisations May 2021 – Oct 2025

What : Led delivery and optimisation of large-scale programmatic advertising campaigns across multiple platforms.

How : Analysed performance data, diagnosed system-level inefficiencies, implemented optimisation strategies, and coordinated across technical platforms and commercial stakeholders under tight deadlines.

Result : Consistently surpassed client KPIs, improved campaign efficiency and audience quality, reduced operational inefficiencies, and delivered measurable improvements in ROI through actionable insights.

SKILLS

- › **Mechanical & CAD :** Fusion 360, Inventor, SolidWorks (basic)
- › **Electronics :** Arduino, embedded C/C++, sensors, LEDs
- › **Manufacturing :** FDM 3D printing, laser cutting
- › **Analysis :** MATLAB, Excel
- › **Programming :** Python (foundational)

Affiliations : Bristol Racing, HyPower Rocket Society, Engineers Without Borders, Academic Research Society, IMechE Student Affiliate

EDUCATION

University of Bristol — Mechanical with Electrical Engineering

2029

Digital Marketing Bootcamp — Multiverse

Codecademy — Python (2, 3)